

Engineering Physics — Electromagnetics & Optics

Course Code: 25B11PH111 / 25B11PH112 • Department of Physics • JUIT Solan

1. Maxwell's Field Equations in Differential & Integral Forms

- Gauss's Law for Electrostatics: $\nabla \cdot \mathbf{E} = \rho / \epsilon_0$
- Gauss's Law for Magnetism: $\nabla \cdot \mathbf{B} = 0$ (No magnetic monopoles)
- Faraday's Law of Induction: $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$
- Ampere-Maxwell Law: $\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 (\partial \mathbf{E} / \partial t)$ [Displacement Current]

Principle of Lasers: Stimulated absorption, spontaneous emission, stimulated emission, and population inversion.

2. Lasers & Fiber Optics

Ruby Laser (3-level system) and Helium-Neon Laser (4-level system, 632.8 nm red beam).

Optical Fibers: Total internal reflection, acceptance angle $\theta_0 = \arcsin(\sqrt{n_1^2 - n_2^2})$, numerical aperture (NA).